Confidence Intervals and Hypothesis Testing

Chapters 8 and 9

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Objectives

- Distinguish a point estimate from an interval estimate, and say what the "95%" in a 95% confidence interval does and does not mean.
- Build and interpret a confidence interval for a mean (with t, and with z when σ is known) and for a proportion.

- Explain how the confidence level, the variability of the data, and the sample size each move the margin of error, and find the sample size a target margin requires.
- State null and alternative hypotheses from a claim, before seeing the data, and choose between a one-tailed and a two-tailed test.
- Define Type I and Type II errors as **false positives** and **false negatives**, with their probabilities α and β.
- Calculate and correctly interpret a p-value for a t-test of a mean and a z-test of a proportion.
- Use the link between a confidence interval and a two-tailed test.
- Calculate and interpret the power of a test, and say how α, the true effect, and n change it.
- Tell statistical significance from practical importance.

Chapter 7 asked: if we knew the population, how would a sample statistic behave? These two chapters turn that question around. A manager holds *one* sample and wants to say something about the population she cannot see. She wants either a range for the unknown value ("average order size is between $118 and $141") or a verdict on a claim ("the new kiosks really are faster than five minutes").

Both answers come from one piece of machinery, built from the standard error of Chapter 7:

$$\text{estimate} \pm (\text{critical value}) \times (\text{standard error}).$$

Read as a range, it is a **confidence interval** (Chapter 8). Solved for n, it tells you **how large a sample** to take (Chapter 8). Rearranged as $(\text{estimate} - \text{claimed value})/\text{standard error}$, it is the **test statistic** of a hypothesis test, and its tail area is the **p-value** (Chapter 9). One engine, three uses. That is why we can cover two chapters in one week, and it is the thread to hold on to.

As before, you will not compute these by hand on an exam. You will be handed output, from Excel or from code, and asked what it means.

Part I: Confidence Intervals (Chapter 8)

Point estimates and interval estimates

A **point estimate** is a single number computed from a sample to estimate a population parameter: the sample mean $\bar{x}$ for μ, the sample proportion $\hat{p}$ for p.

An **interval estimate**, or **confidence interval**, is a range of values, computed from the sample, constructed by a method that captures the parameter in a stated share of samples, the **confidence level**, such as 95%.

A point estimate is the best single guess, and it is almost certainly wrong: Chapter 7 showed that $\bar{x}$ lands above or below μ by a sampling error that changes from sample to sample. A point estimate says nothing about how wrong it is likely to be. The interval does, and its half-width has a name.

The **margin of error** is the distance from the point estimate to either end of a confidence interval:

$$\text{margin of error} = (\text{critical value}) \times (\text{standard error}), \qquad \text{interval} = \text{estimate} \pm \text{margin of error}\ .$$

The three intervals of this week

Parameter	Estimate	Standard error	Critical value	Excel margin of error
mean, σ known	$\bar{x}$	$\sigma/\sqrt{n}$	$z_{\alpha/2}$	`=CONFIDENCE.NORM(alpha, sigma, n)`
mean, σ unknown (the usual case)	$\bar{x}$	$s/\sqrt{n}$	$t_{\alpha/2,\,n-1}$	`=CONFIDENCE.T(alpha, s, n)`
proportion	$\hat{p} = x/n$	$\sqrt{\hat{p}(1-\hat{p})/n}$	$z_{\alpha/2}$	`=NORM.S.INV(1-alpha/2)*SE`

Here $\alpha = 1 - \text{confidence level}$, so a 95% interval has $\alpha = 0.05$, split as 0.025 in each tail. If the sample is more than 5% of a finite population, multiply the standard error by the finite population correction $\sqrt{(N-n)/(N-1)}$ exactly as in Chapter 7 (Section 8.6 of the text works an example).

When σ is known

Your textbook starts here because the arithmetic is simplest. QVC samples 32 orders: $\bar{x} = \$129.20$, and the population standard deviation is taken to be known, $\sigma = \$40.60$.

confidence	critical value	margin of error	lower	upper	width
90%	1.645	11.805	117.395	141.005	23.611
95%	1.960	14.067	115.133	143.267	28.134
99%	2.576	18.487	110.713	147.687	36.974

> At 95%, we estimate that the average QVC order is between \$115.13 and \$143.27. Asking for 99% confidence widens the interval to \$36.97, against \$23.61 at 90%. **More confidence costs width.** The sample is the same in every row; only the promise changed. (The textbook prints slightly different limits because it rounds the standard error to 7.173 before multiplying.)

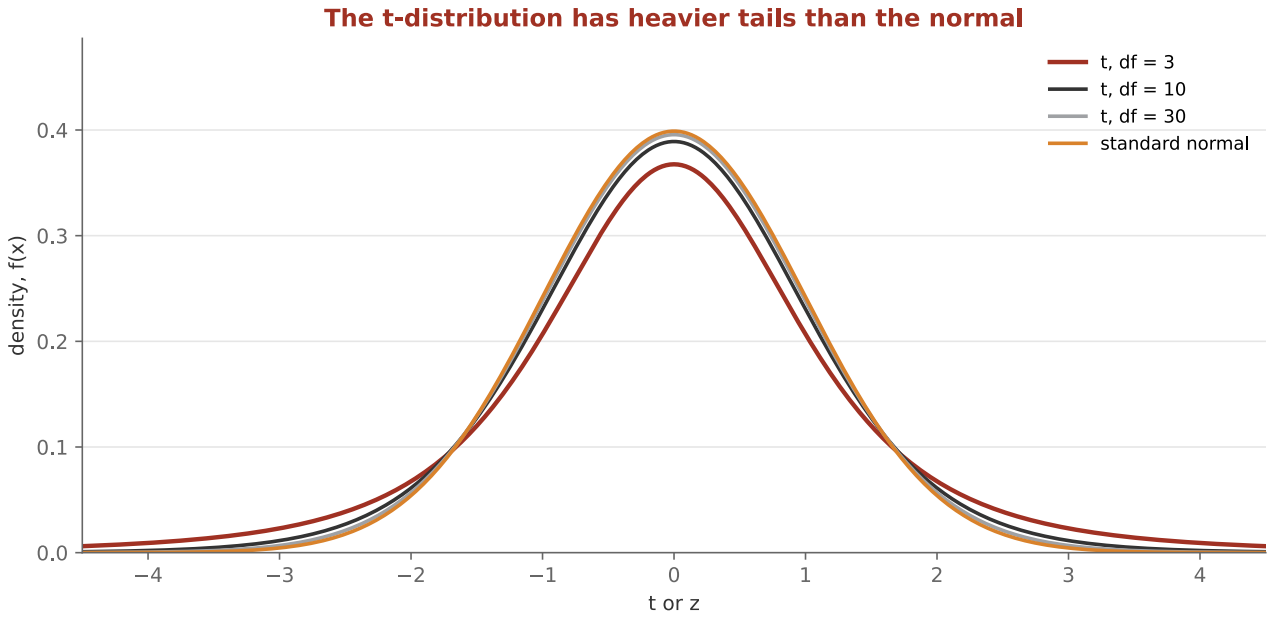
In practice σ is almost never known. If you knew the population's standard deviation you would very likely know its mean too, and there would be nothing to estimate. The realistic case replaces σ with s, and that changes the critical value.

When σ is unknown: the Student's t-distribution

> When the sample standard deviation s replaces σ, the standardized mean
>
> $$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$
>
> follows the **Student's t-distribution** with $n-1$ **degrees of freedom** (df), provided the population is normal. The degrees of freedom count the values free to vary once $\bar{x}$ is fixed: knowing the mean and $n-1$ of the values pins down the last one.

Properties of the t-distribution

- Bell-shaped and symmetric around 0, with total area 1.
- Flatter in the middle and **heavier in the tails** than the normal, because s itself varies from sample to sample and adds uncertainty.
- A family of curves, one for each df. As df grows, s becomes a reliable stand-in for σ and t approaches the standard normal.
- So the critical t is always **larger** than the critical z for the same confidence, and t-intervals are wider.



	90%	95%	99%
t, df = 5	2.015	2.571	4.032
t, df = 10	1.812	2.228	3.169
t, df = 20	1.725	2.086	2.845
t, df = 30	1.697	2.042	2.750
t, df = 60	1.671	2.000	2.660
t, df = 100	1.660	1.984	2.626
z (normal)	1.645	1.960	2.576

At 95% confidence the critical value is 2.228 with 10 df, 2.042 with 30, and 1.984 with 100, against 1.960 for z. With small samples the difference matters; past a hundred or so degrees of freedom it is cosmetic. Excel's `=T.INV.2T(0.05, df)` returns these values.

William Sealy Gosset worked out this distribution in 1908 while testing barley and hops for the Guinness brewery in Dublin, where samples were small and expensive. Guinness did not let its staff publish under their own names, so he signed the paper "Student." The most-used small-sample method in statistics came out of quality control in a business.

Application: Big Ten faculty salaries

In Chapter 7 we met the 215 economics faculty at Big Ten universities in 1999. Because we have the whole population, we know the true mean salary, $93,457, and can check our interval against it. Pretend we can afford to look at only 30 of them.

	value
n	30
s (sample sd)	32565.0357

	value
estimate (x-bar)	105090.3
standard error	5945.5349
distribution	t, df = 29
critical value	2.0452
margin of error	12159.9842
lower	92930.3158
upper	117250.2842
Excel margin of error	=CONFIDENCE.T(0.05, 32565, 30)

$$\bar{x} \pm t_{0.025,\,29}\frac{s}{\sqrt{n}} = 105,090 \pm 2.045 \times \frac{32,565}{\sqrt{30}} = 105,090 \pm 12,160$$

We estimate, with 95% confidence, that the mean salary of the 215 faculty is between $92,930 and $117,250. The true mean, $93,457, is inside this interval. This particular sample ran high, by $11,634, and the interval still reached the truth with $526 to spare. Whether a given interval catches the truth is a matter of luck. How often the method catches it is not, as the next section shows.

The textbook's Dr. Gregg samples 18 patients and finds a mean of 100.3 with a sample standard deviation of 16.6. Find the 95% confidence interval for the population mean, and write the Excel formula for its margin of error. What must be true of the population for this interval to be trustworthy with only 18 observations?

i Solution

	value
estimate (x-bar)	100.3
standard error	3.9127
distribution	t, df = 17
critical value	2.1098
margin of error	8.255
lower	92.045
upper	108.555
Excel margin of error	=CONFIDENCE.T(0.05, 16.6, 18)

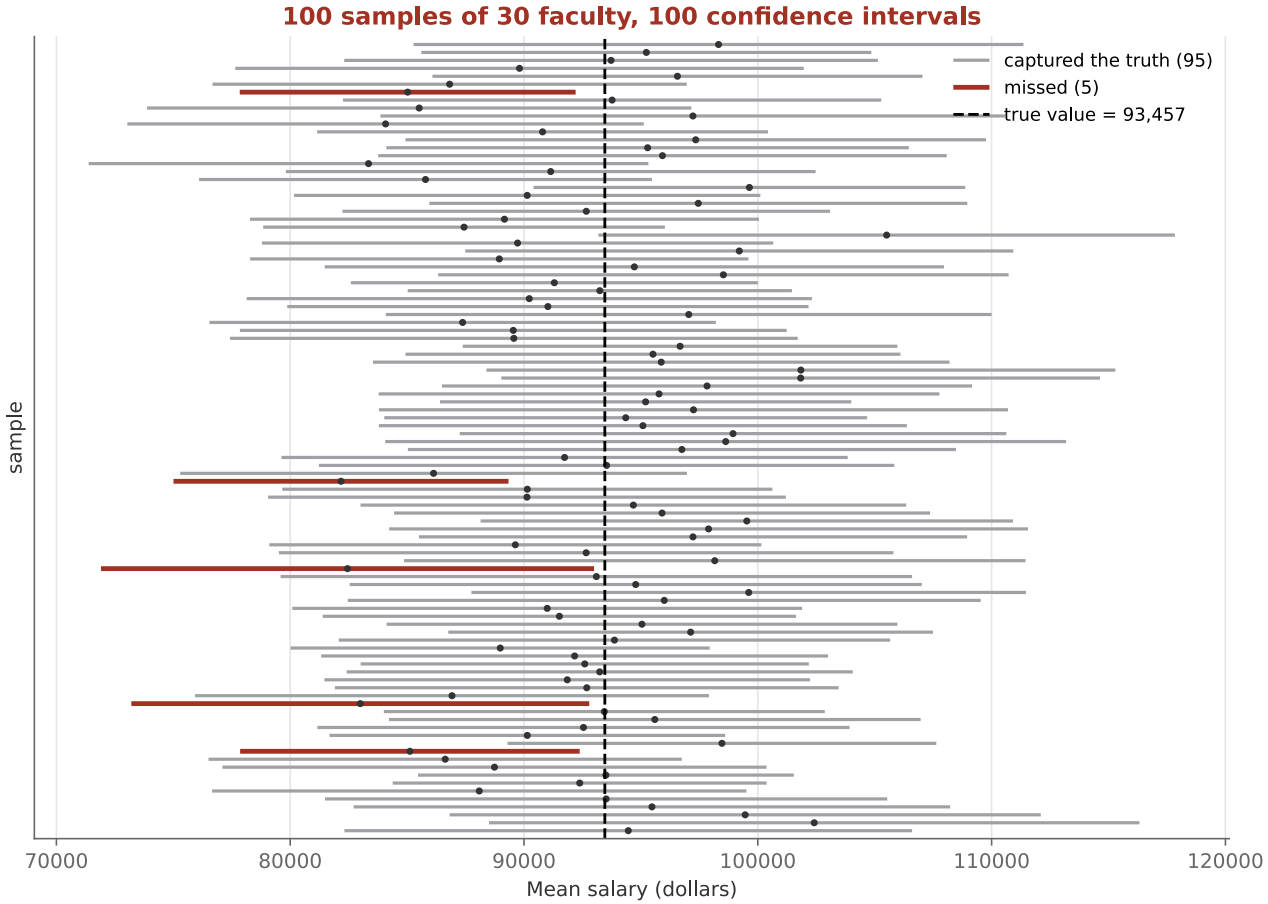
The interval is 92.05 to 108.55, with $t_{0.025,17} = 2.110$. The margin of error is `=CONFIDENCE.T(0.05, 16.6, 18)`. With $n = 18$ the Central Limit Theorem cannot be relied on, so the population itself must be close to normal.

What “95% confident” means

A confidence interval is not a probability statement about μ. Once the sample is drawn, the interval is fixed and so is μ: the interval either contains μ or it does not. There is nothing left to be random. It is wrong to say “there is a 95% probability that μ is between \$92,930 and \$117,250.”

The **confidence level** describes the **method**, not any one interval: if we drew sample after sample and built an interval from each in the same way, 95% of those intervals would capture μ. The share that miss is α. “We are 95% confident” is shorthand for “this interval came from a method that works 95% of the time.”

We can watch the method work, because for the faculty we know μ. Each horizontal line below is the 95% t-interval from a fresh sample of 30 (drawn with replacement, so the population behaves as if it were infinite).



95 of these 100 intervals captured the true mean, and 5 missed (in scarlet). Nothing about a miss looks different from the inside: each was computed correctly from a perfectly good random sample. You never learn which kind you are holding. That is the price of estimating from a sample, and the confidence level is how you choose to pay it.

Repeat the experiment 100,000 times and the share that capture the truth settles down. From a normal population it is 95.1%, as promised. From the faculty salaries it is 94.6%, a little short of 95%, because salaries are right-skewed and 30 observations do not fully cure that. Keep that number in mind for the section after next.

The margin of error and its three levers

The margin of error is $(\text{critical value}) \times s/\sqrt{n}$. Three things move it, and a manager controls two of them.

1. **The confidence level.** Higher confidence means a larger critical value and a wider interval (the QVC table above).
2. **The variability of the data, s.** Noisier data, wider interval. You rarely control this, though a better measurement process or a stratified design (Chapter 7) can reduce it.
3. **The sample size, n.** Larger samples, narrower intervals, but through $\sqrt{n}$: to halve the margin you need about **four times** the data.

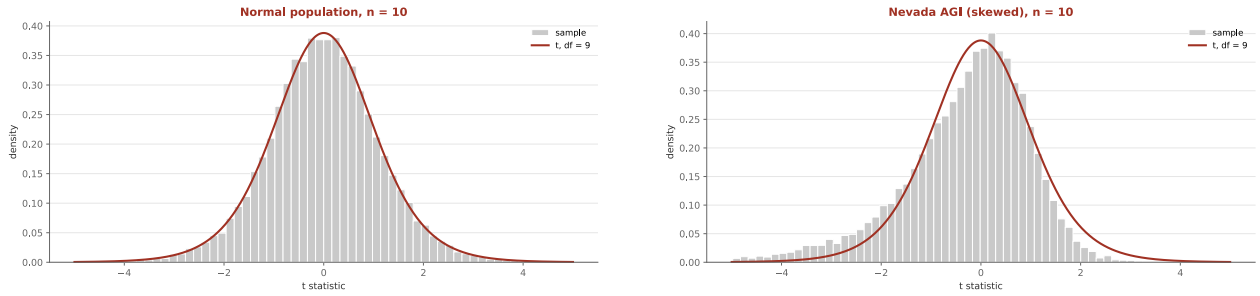
n	standard error	critical value	margin of error
25	5900.0	2.064	12177.0
50	4171.9	2.010	8383.8
100	2950.0	1.984	5853.4
200	2086.0	1.972	4113.4
400	1475.0	1.966	2899.7
1600	737.5	1.961	1446.6

With a standard deviation like the faculty salaries' (about \$29,500), a sample of 100 gives a 95% margin of error of \$5,853; quadrupling to 400 gives \$2,900, a little better than half because the critical t also falls. Precision is expensive. The next section turns this around and asks how large a sample a target margin requires.

When is the t-interval trustworthy?

The t-interval assumes the **population is normal**. The assumption matters most when n is small. As n grows, the Central Limit Theorem makes $\bar{x}$ close to normal anyway, and the t-procedure becomes reliable for most populations, **but how large is "large" depends on how skewed the population is.**

The textbook's rule of thumb is $n \geq 30$. The honest version is a simulation. Below, we draw thousands of samples from a population whose mean we know, compute the t-statistic for each, and compare its distribution with the t-curve it is supposed to follow. On the left the population is normal. On the right it is the Nevada adjusted gross income file from Chapter 7, which is strongly right-skewed.



The pictures hint at the problem; the tail counts measure it. These are the actual error rates of 5% one-tailed tests, which is the same as asking how often a one-sided 95% bound would miss.

	lower-tailed test rejects	upper-tailed test rejects	two-sided test rejects	nominal alpha
normal, n = 10	0.049	0.050	0.050	0.05
normal, n = 30	0.049	0.050	0.046	0.05
Nevada AGI, n = 10	0.117	0.018	0.086	0.05
Nevada AGI, n = 30	0.089	0.026	0.067	0.05
Nevada AGI, n = 100	0.071	0.037	0.060	0.05
Nevada AGI, n = 400	0.059	0.040	0.051	0.05

With a normal population the t-procedure does exactly what it promises, even at $n = 10$. With the skewed income data at $n = 30$, a test that is supposed to be wrong 5% of the time in the lower tail is wrong 8.9% of the time, and in the upper tail only 2.6%. The errors pile up on one side. Even at $n = 400$ the two tails are 5.9% and 4.0%. Two-tailed procedures fare better, because the two errors partly cancel.

"$n \geq 30$" is a starting point, not a guarantee. For incomes, sales, claims, wait times and anything else with a long right tail, look at the data before trusting a t-interval or t-test, and be most careful with one-tailed tests. Your textbook suggests that a larger sample helps "where the t-distribution can be approximated by the normal distribution." The reason a larger sample helps is the Central Limit Theorem acting on $\bar{x}$, not t turning into z. Chapter 12 introduces a formal check of normality.

A confidence interval for a proportion

For a population proportion, with x successes in a sample of n and $\hat{p} = x/n$,

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

The normal approximation needs enough successes and failures: $n\hat{p} \geq 5$ and $n(1-\hat{p}) \geq 5$ (many texts ask for 10; the stricter rule is safer).

The textbook surveys 175 people, of whom 116 have the characteristic of interest, and asks for a 99% interval.

	value
successes x	116
n	175
estimate (p-hat)	0.6629
n p-hat (want >= 5)	116.0
n (1 - p-hat) (want >= 5)	59.0
standard error	0.0357
critical value z	2.5758
margin of error	0.092
lower	0.5708
upper	0.7549
Excel critical value	=NORM.S.INV(0.995)

The sample proportion is 0.663, and with 99% confidence the population proportion is between 0.571 and 0.755, a margin of error of 0.092, or about 9 percentage points. News reports of polls quote this margin, almost always at 95%.

The standard error here uses $\hat{p}$, because the interval makes no claim about p and $\hat{p}$ is the best we have. When we *test* a claimed proportion in Part II, the standard error will use the claimed value instead. Students mix these up more than anything else in the proportion material.

How large a sample do you need?

This is the most practical result in Chapter 8. Set the margin of error you can live with, E, and solve for n.

For a mean: $n = \left(\frac{z_{\alpha/2}\,\sigma}{E}\right)^2$.

For a proportion: $n = \frac{z_{\alpha/2}^2\,p(1-p)}{E^2}$.

Always round up. Rounding down gives a margin slightly larger than you asked for. σ and p are *planning values* from a pilot study or past data. With no information about p, use $p = 0.5$, which makes $p(1-p)$ as large as it can be, so the sample is never too small.

	(z sigma / E)^2	n (round UP)	planning p	sigma (planning value)	target margin E	z	z^2 p (1-p) / E^2
AT&T: mean, sigma = 400, E = 75, 95%	109.268	110.0		400.0	75.00	1.960	
proportion, p = 0.12, E = 0.05, 98%		229.0	0.12		0.05	2.326	228.598
proportion, p = 0.5, E = 0.05, 98%		542.0	0.5		0.05	2.326	541.189

AT&T needs 110 customers: the formula gives 109.27, and 109 would fall just short. For the proportion, a prior estimate of 12% requires 229; with no prior information (p = 0.5) it takes 542. Knowing something about p in advance saves more than half the sample.

The textbook reports 230 and 543. It uses $z = 2.33$ from its table; Excel's `=NORM.S.INV(0.99)` gives 2.326, and the extra precision moves each answer by one. Either is acceptable. Just be consistent.

margin of error E	n
0.08	151
0.04	601
0.02	2401
0.01	9604

Halving the margin of error quadruples the sample. This one table explains why national polls settle on about 1,000 respondents (a margin near three points) and almost never pay for the 10,000 that would buy one point.

A Las Vegas resort wants to estimate the average amount a convention attendee spends on property, to within $15 with 95% confidence. Last year's survey suggests a standard deviation of about $110. **(a)** How many attendees should it survey? **(b)** It also wants the share who would return next year, to within 4 points at 95%, with no prior estimate. How many for that? **(c)** One survey will ask both questions. How many responses does it need?

Solution

	(z sigma / E)^2	n (round UP)	planning p	sigma (planning value)	target margin E	z	z^2 p (1-p) / E^2
(a) mean	206.585	207.0		110.0	15.00	1.96	
(b) proportion		601.0	0.5		0.04	1.96	600.228

(a) 207. **(b)** 601. **(c)** The larger of the two, 601, so that both targets are met. And in practice more than that, since not everyone asked will answer, a nonsampling problem (Chapter 7) that no formula fixes.

Part II: Hypothesis Testing (Chapter 9)

The logic of a test

A **hypothesis** is a claim about a population parameter, such as a mean or a proportion.

The **null hypothesis**, H_0, is the status quo or the claim being tested. It always contains the equality: $=$, $\leq$, or $\geq$ a specific value.

The **alternative hypothesis**, H_1, is what we would conclude if the data are strong enough evidence against H_0. It never contains the equality: $\neq$, $<$, or $>$.

A hypothesis test is a trial. H_0 is the defendant, presumed innocent. The data are the evidence. The verdict is **reject H_0** (guilty: the evidence is strong enough) or **do not reject H_0** (not guilty: it is not). A jury never declares a defendant *innocent*, only *not guilty*, because a lack of evidence is not proof of innocence.

We never "accept" H_0. Failing to reject it means the data are **consistent with** H_0, not that they show it is true. The same data are usually consistent with many other values too. Your textbook occasionally says a sample "supports the original assumption" after a failure to reject. Read that as "fails to contradict."

Stating the hypotheses

Three claims, three setups.

Time on a web page. A study reported in the *Wall Street Journal* (Does the Internet Make You Smarter? and Does the Internet Make You Dumber?) says the average American looks at a web page for 56 seconds. A student wants to know whether UNLV students are *different*, in either direction:

$$H_0 : \mu = 56 \qquad H_1 : \mu \neq 56 \qquad \text{(two-tailed)}$$

Light bulb life. A lab wants evidence that a new bulb *lasts longer* than the industry's 8,000 hours. The claim needing evidence goes in H_1:

$$H_0 : \mu \leq 8{,}000 \qquad H_1 : \mu > 8{,}000 \qquad \text{(one-tailed, upper)}$$

An 18-game NFL season. The commissioner says a majority of fans favor it. The players' union, which opposes it, wants evidence that the commissioner is wrong:

$$H_0 : p \geq 0.5 \qquad H_1 : p < 0.5 \qquad \text{(one-tailed, lower)}$$

A **two-tailed test** has $H_1 :\neq$ and looks for a difference in either direction. A **one-tailed test** has $H_1 :<$ or $H_1 :>$ and looks in one direction only.

Choose the hypotheses and the tail before you look at the data. H_1 is the claim that *requires evidence*, not the answer you are hoping for. A one-tailed p-value is half the two-tailed one, so choosing the tail after seeing which way the data point turns a two-tailed p of 0.08 into a one-tailed 0.04 and an "insignificant" result into a "significant" one. That is not analysis; it is a thumb on the scale. When in doubt, test two-sided.

Two ways to be wrong: false positives and false negatives

Because the verdict rests on a sample, it can be wrong in two ways, and the two mistakes have very different costs.

In the language of testing, a **positive** result is a test that **rejects H_0**: it reports that it found something, an effect, a difference, a disease, a fraud, a benefit. A **negative** result is a test that does **not** reject H_0: it reports that it found nothing.

A **Type I error** is a **false positive**: rejecting H_0 when H_0 is actually true. The test reports an effect that is not there. Its probability is α, the **significance level**, which we choose:

$$\alpha = P(\text{reject } \ H_0 \mid H_0 \ \text{ is true}).$$

A **Type II error** is a **false negative**: failing to reject H_0 when H_0 is actually false. The test misses an effect that is there. Its probability is β:

$$\beta = P(\text{do not reject } \ H_0 \mid H_0 \ \text{ is false}).$$

The **power** of a test is the probability of a **true positive**, correctly rejecting a false H_0: $\text{power} = 1 - \beta$.

	H0 is actually TRUE (no effect)	H0 is actually FALSE (real effect)
Reject H0 (test is 'positive')	Type I error = FALSE POSITIVE (probability alpha)	Correct: true positive (probability 1 - beta = power)
Do not reject H0 (test is 'negative')	Correct: true negative (probability 1 - alpha)	Type II error = FALSE NEGATIVE (probability beta)

	H_0 actually true (nothing there)	H_0 actually false (something there)
Courtroom (H_0: innocent)	convict an innocent person: **false positive, Type I**	acquit a guilty person: **false negative, Type II**
Smoke alarm (H_0: no fire)	alarm sounds, no fire: **false positive, Type I**	fire, no alarm: **false negative, Type II**
Spam filter (H_0: real mail)	real mail sent to spam: **false positive, Type I**	spam reaches the inbox: **false negative, Type II**
Audit (H_0: no fraud)	honest account flagged: **false positive, Type I**	fraud goes undetected: **false negative, Type II**

A memory aid: a Type I error is **crying wolf** (an alarm with no wolf); a Type II error is **missing the wolf**. "Positive" never means "good"; it means the test said *yes, something is here*. A positive cancer screen is bad news and a positive drug trial is good news; a *false* positive is a wrong "yes" either way.

Your textbook gets these labels backwards in one place. The Focus on Analytics box on the FDA's drug approvals, at the end of Section 9.5, says "a Type I error (false negative)" and "a Type II error (false positive)." It is the other way around. With H_0: *the drug has no benefit*,

- approving a drug that does not work is rejecting a true H_0: a **Type I error, a false positive**;
- refusing a drug that does work is failing to reject a false H_0: a **Type II error, a false negative**.

The box's own definitions and its decision figure are correct; only the two parenthetical labels are swapped. Use the definitions above on exams.

The FDA example is still worth reading for its point about costs. A false positive (a harmful or useless drug approved) produces identifiable victims and headlines, so regulators guard hard against it with a small α. A false negative (a beneficial drug refused) produces victims nobody can name, people who would have been helped, so it gets less attention. A small α buys protection against the visible error at the price of a larger β, more of the invisible one. Which error is worse is a business or policy judgment, not a statistical one, and it should drive the choice of α.

α is $P(\text{reject} \mid H_0 \text{ true})$, the false positive rate *among situations where nothing is there*. It is **not** the probability that a rejection is a false positive, $P(H_0 \text{ true} \mid \text{reject})$. That second number also depends on how often H_0 is true to begin with: the base-rate problem of the screening-test example in Chapter 4. If a company tests 100 marketing ideas and only a few really work, a good share of its "significant" results will be false positives even at $\alpha = 0.05$.

In quality control, a Type I error (rejecting a good batch) is sometimes called the **producer's risk** and a Type II error (accepting a bad batch) the **consumer's risk**. The structure is the same.

α and β pull against each other. For a fixed sample size, making it harder to cry wolf makes it easier to miss the wolf. The only way to reduce both at once is a larger sample. We return to this with numbers at the end of the chapter.

The p-value

The **p-value** is the probability, **computed assuming H_0 is true**, of getting a test statistic at least as extreme as the one observed, in the direction(s) of H_1.

Decision rule: reject H_0 if $p \le \alpha$; otherwise do not reject.

A small p-value says: *if* H_0 were true, data like ours would be rare. Either we saw something rare, or H_0 is false. Below α, we bet on the second. This is the same reasoning we used to judge the commute claim in Chapter 7. The p-value is simply that tail probability, given a name.

What the p-value is not.

- It is **not** the probability that H_0 is true. It is computed by *assuming* H_0 is true, so it cannot also measure whether it is.
- It is **not** "the confidence you have that your null hypothesis is correct," as your textbook says in Section 9.2. A p-value of 0.30 does not mean we are 30% sure of H_0.
- It is **not** the size or importance of an effect. A tiny, worthless effect can have a tiny p-value if the sample is big enough (see the end of this note).

Your textbook adds that a p-value "can be computed directly only for z." That was a limit of printed tables, not of the method. Excel's `T.DIST`, `T.DIST.RT`, and `T.DIST.2T` give exact t p-values, and the book itself uses them in Section 9.3.

The six steps

1. State H_0 and H_1 (before seeing the data).
2. Choose α, by weighing the cost of a false positive against a false negative.
3. Identify the test statistic and its distribution under H_0 (and, if you use them, the critical values).
4. Compute the test statistic from the sample.
5. Compute the p-value and compare it with α (equivalently, compare the statistic with the critical value).
6. State the conclusion **in the words of the problem**, including the direction, and never "accept H_0."

For a mean with σ unknown, the test statistic is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \mathrm{df} = n - 1,$$

where μ_0 is the value **claimed in H_0**. The numerator is not $\bar{x} - \mu$: we do not know μ, and the whole calculation asks how far $\bar{x}$ is from what H_0 says.

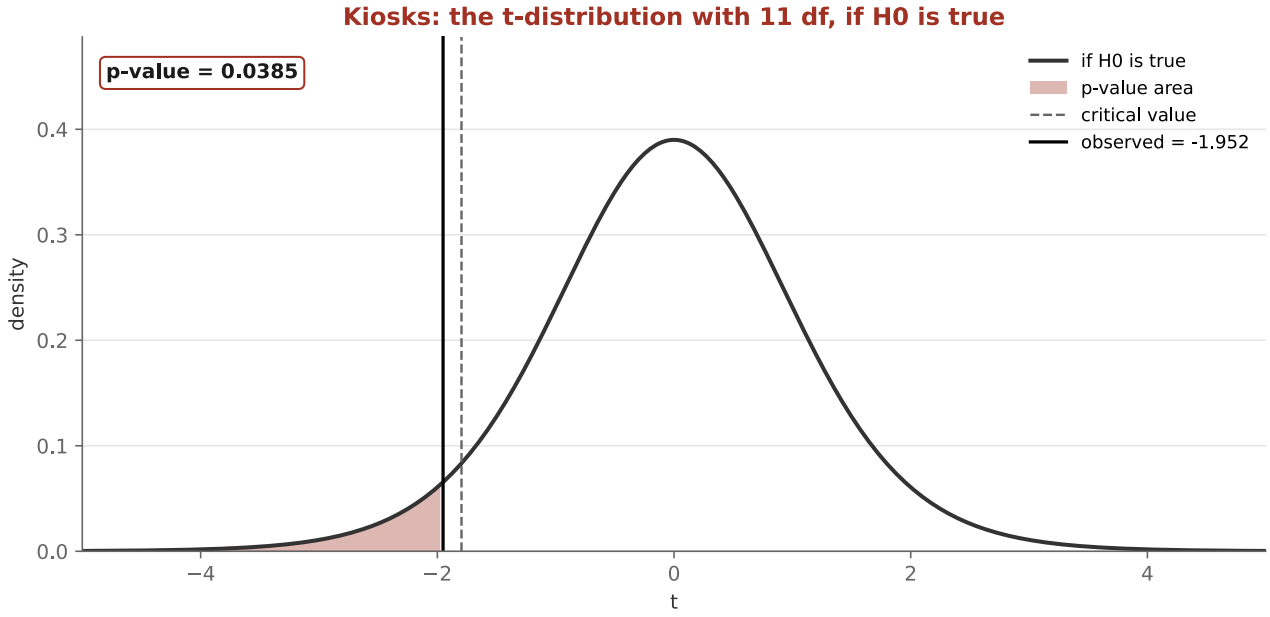
Application: a one-tailed t-test

Target pilots self-service checkout kiosks. A cashier checkout averages 5 minutes, and Target will adopt the kiosks only if they are faster. Twelve kiosk checkouts average $\bar{x} = 4.2$ minutes with $s = 1.42$. Test at $\alpha = 0.05$.

$$H_0 : \mu \ge 5 \qquad H_1 : \mu < 5$$

	value
H0 value mu0	5
alternative	less
x-bar	4.2
s	1.42
n	12
standard error s/sqrt(n)	0.4099
test statistic t	−1.9516

	value
df	11
critical value	−1.7959
p-value	0.0385
alpha	0.05
reject H0?	True
Excel p-value	=T.DIST(-1.9516, 11, TRUE)



$t = -1.95$, and the p-value is 0.0385 (Excel: =T.DIST(-1.9516, 11, TRUE)). If kiosks really took 5 minutes or more on average, a sample of 12 averaging 4.2 or less would turn up only about 3.8% of the time. That is below $\alpha = 0.05$, so we reject H_0: **the data are evidence that average kiosk checkout time is less than 5 minutes.** The statistic is also past the critical value, −1.796; the two comparisons always agree.

Two cautions before Target rips out the registers. With $n = 12$ the test relies on checkout times being roughly normal, and checkout times are usually right-skewed, exactly the case where a one-tailed test misbehaves (the simulation above). And "less than 5 minutes" could mean 4.9: the next step is a confidence interval for how much faster.

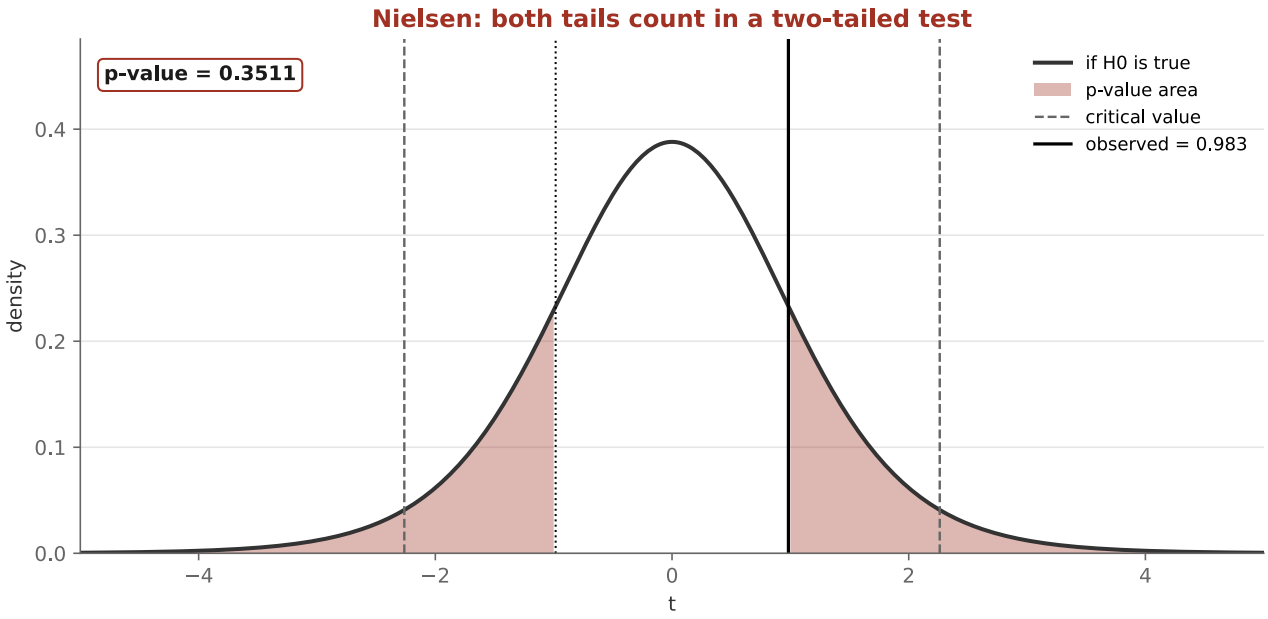
Application: a two-tailed t-test

Nielsen says the average American watches 34.5 hours of television a week. Ten people average $\bar{x} = 39.6$ hours with $s = 16.4$. Is the claim wrong, in either direction? Use $\alpha = 0.05$.

$$H_0 : \mu = 34.5 \qquad H_1 : \mu \neq 34.5$$

	value
H0 value mu0	34.5
alternative	two-sided

	value
x-bar	39.6
s	16.4
n	10
standard error s/sqrt(n)	5.1861
test statistic t	0.9834
df	9
critical value	+/- 2.2622
p-value	0.3511
alpha	0.05
reject H0?	False
Excel p-value	=T.DIST.2T(0.9834, 9)



$t = 0.98$. In a two-tailed test, a result as far *below* 34.5 would be just as much evidence against H_0, so the p-value counts both tails: 0.351 (Excel: =T.DIST.2T(0.9834, 9), which needs the *absolute value* of t). That is far above 0.05, so we do not reject H_0: **the sample is not evidence that Americans' average viewing differs from 34.5 hours.**

It is *not* evidence that the average **is** 34.5. Ten people with a standard deviation of 16 hours could hardly contradict any claim within several hours of 39.6. The next section makes "several hours" precise.

Tests and confidence intervals are the same calculation

A two-tailed test at significance level α rejects $H_0 : \mu = \mu_0$ **exactly when** μ_0 lies outside the $(1 - \alpha)$ confidence interval. The confidence interval is the set of all claimed values the test would not reject.

```
95% interval: 27.87 to 51.33
```

claimed mu0	inside the interval?	two-sided p-value	test rejects H0?
20.0	False	0.0044	True
25.0	False	0.0202	True
30.0	True	0.0972	False
34.5	True	0.3511	False
50.0	True	0.0759	False
55.0	False	0.0157	True

> The 95% interval runs from 27.9 to 51.3 hours. Any claim in that range survives a two-tailed test at $\alpha = 0.05$; any claim outside it is rejected. Notice how wide that is: the same ten people "fail to reject" 30 hours and 50 hours. That is why failing to reject 34.5 says so little.
>
> This is the most useful fact in these two chapters. **The interval tells you everything the two-tailed test does, and more**: not only whether the claim survives, but which other values would. When you report a test, report the interval too. (Your textbook does this with an iPhone price claim of \$811: the 95% interval $797.8 \pm 1.96(5.41)$ excludes 811, so the claim is rejected.)

> The match is exact only for a **two-tailed** test with $\text{confidence} = 1 - \alpha$. A one-tailed test at $\alpha = 0.05$ matches a one-sided 95% bound, or equivalently a two-sided **90%** interval.

Application: light bulbs, with the truth known

The old version of this note simulated a light-bulb study so that we would know the right answer. A new bulb's life in hours is normal with mean 8,050 and standard deviation 225, so it *does* last longer than 8,000, and H_0 is false. The lab tests 100 bulbs.

	value
H0 value mu0	8000
alternative	greater
x-bar	8023.2219
s	191.6617
n	100
standard error s/sqrt(n)	19.1662
test statistic t	1.2116
df	99
critical value	1.6604
p-value	0.1143
alpha	0.05
reject H0?	False

	value
Excel p-value	=T.DIST.RT(1.2116, 99)

The sample averages 8,023.2 hours, $t = 1.21$, p-value 0.1143. We do not reject H0, and that is the WRONG decision: we built the population so that the bulbs really do last longer. This is a Type II error, a false negative, caught in the act. Nothing was done incorrectly. The sample simply came in low, and 100 bulbs were not enough to be sure of detecting a 50-hour improvement.

How often would this study find the improvement? Run it 20,000 times on fresh samples: it rejects H_0 71.3% of the time. That is the study's **power**. About 29% of the time, a perfectly good study of a genuinely better bulb would report "no evidence of improvement."

Testing a proportion

To test $H_0 : p = p_0$ (or $\leq, \geq$), with $\hat{p} = x/n$,

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}},$$

valid when $np_0 \geq 5$ and $n(1 - p_0) \geq 5$. The p-value comes from the standard normal.

The standard error uses p_0, the value claimed in H_0, not $\hat{p}$. The test asks how surprising $\hat{p}$ would be *if* H_0 *were true*, so it uses the spread that H_0 implies. The confidence interval, which claims nothing, uses $\hat{p}$.

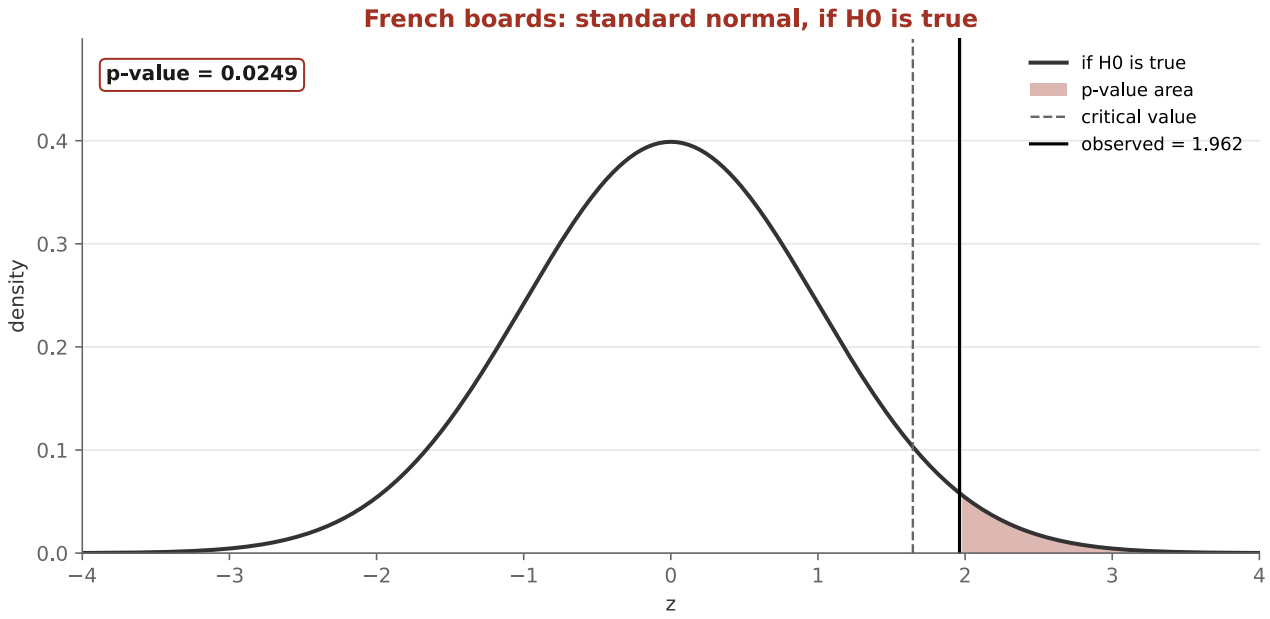
Application: women on French corporate boards

France required large companies to move toward 40% women on their boards. In 2017, 34% of board members were women. Has the share risen? A sample of 240 board members finds 96 women. Test at $\alpha = 0.05$.

$$H_0 : p \leq 0.34 \qquad H_1 : p > 0.34$$

	value
H0 value p0	0.34
alternative	greater
successes x	96
n	240
p-hat	0.4
n p0 (want >= 5)	81.6
n (1 - p0) (want >= 5)	158.4
standard error sqrt(p0 (1-p0) / n)	0.0306
test statistic z	1.9622
critical value	1.6449
p-value	0.0249
alpha	0.05
reject H0?	True

	value
Excel p-value	=1 - NORM.S.DIST(1.9622, TRUE)



$\hat{p} = 0.40$, $z = 1.96$, and the p-value is 0.0249. Below 0.05, so we reject H_0: **the data are evidence that the share of women on large French boards has risen above 34%.** The standard error was 0.0306, computed from $p_0 = 0.34$; using $\hat{p} = 0.40$ would have given 0.0316, a different (and wrong) z.

Finishing the NFL example. Suppose the players' union polls 400 randomly chosen fans and 186 favor an 18-game season. (These numbers are illustrative.) **(a)** State H_0 and H_1. **(b)** Test at $\alpha = 0.05$ and state the conclusion in words. **(c)** Find a 95% confidence interval for the share of fans in favor. Which standard error did you use in (b), and which in (c)?

i Solution

	value
H0 value p0	0.5
alternative	less
successes x	186
n	400
p-hat	0.465
n p0 (want >= 5)	200
n (1 - p0) (want >= 5)	200
standard error sqrt(p0 (1-p0) / n)	0.025
test statistic z	−1.4
critical value	−1.6449
p-value	0.0808
alpha	0.05
reject H0?	False
Excel p-value	=NORM.S.DIST(-1.4000, TRUE)

(a) $H_0 : p \geq 0.5$ (the commissioner's claim), $H_1 : p < 0.5$ (what the union wants evidence for). **(b)** $\hat{p} = 0.465$, $z = -1.40$, p-value 0.0808. Do not reject H0: although fewer than half of this sample favor the change, the poll is not strong enough evidence that fewer than half of ALL fans do. **(c)** 0.416 to 0.514. The test used $p_0 = 0.5$ in its standard error; the interval used $\hat{p} = 0.465$.

When σ is known: the z-test

When the population standard deviation is known, the test statistic is $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$ and the p-value comes from the standard normal. Everything else is identical. Your textbook works several examples this way in Section 9.2; read them for the steps, and use the t-test in practice.

	value
H0 value mu0	8000
alternative	greater
x-bar	8120
sigma (known)	500
n	36
standard error sigma/sqrt(n)	83.3333
test statistic z	1.44
critical value	1.6449
p-value	0.0749
alpha	0.05

	value
reject H0?	False
Excel p-value	=1 - NORM.S.DIST(1.4400, TRUE)

The textbook's Edalight example: 36 new compact fluorescent bulbs average 8,120 hours, and $\sigma = 500$ hours is taken as known. Testing $H_0 : \mu \leq 8,000$ against $H_1 : \mu > 8,000$ gives $z = 1.44$ and p-value 0.0749, which is above $\alpha = 0.05$: not enough evidence that the new bulb lasts longer than 8,000 hours.

Excel's `Z.TEST(range, mu0, sigma)` returns the **upper-tail** p-value only. For a lower-tailed test use `1 - Z.TEST(...)`; for a two-tailed test, `2 * MIN(Z.TEST(...), 1 - Z.TEST(...))`. (The textbook's description of its arguments calls σ "the population mean"; it is the population standard deviation.)

Type II errors and power

α is chosen. β is not: it depends on how wrong H_0 actually is, on α, and on n. To compute it we must pick a specific "true" value, because "H_0 is false" covers infinitely many possibilities.

Application: the Nissan Leaf battery

Nissan's Leaf averages 151 miles per charge. A new battery is claimed to do better. Nissan will test 50 cars, with $\sigma = 12$ miles known and $\alpha = 0.05$:

$$H_0 : \mu \leq 151 \qquad H_1 : \mu > 151.$$

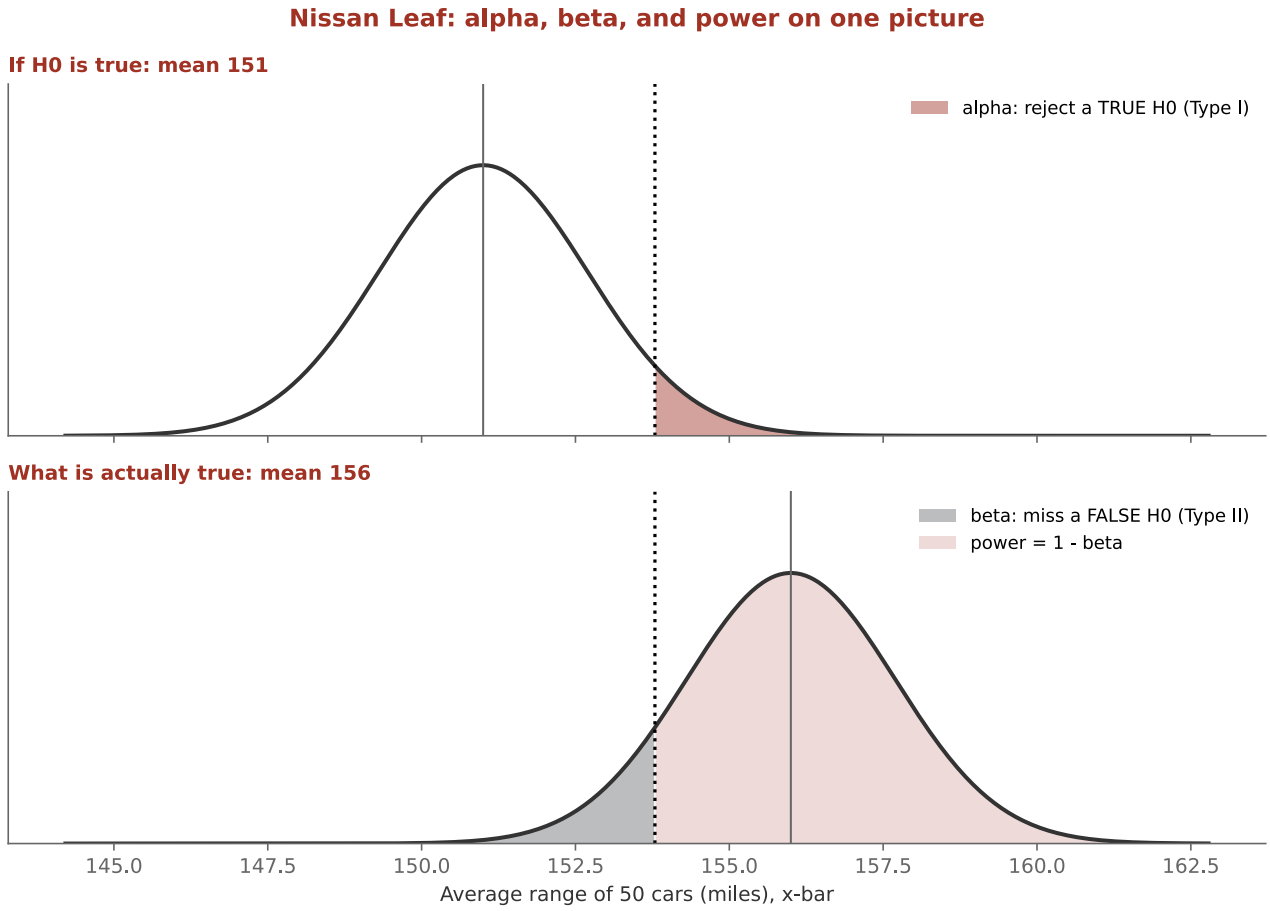
The **critical sample mean** $\bar{x}_\alpha$ is the value of $\bar{x}$ at the edge of the rejection region. For an upper-tailed test,

$$\bar{x}_\alpha = \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}},$$

(with $-$ for a lower-tailed test). If the true mean is μ_1,

$$\beta = \mathbb{P}(\bar{x} < \bar{x}_\alpha \mid \mu = \mu_1), \qquad \text{power} = 1 - \beta.$$

	value
mu0 (H0)	151
true mean	156
sigma	12
n	50
alpha	0.05
standard error	1.6971
critical x-bar	153.7914
beta = P(Type II error)	0.0966
power = 1 - beta	0.9034



Nissan rejects H_0 if the 50 cars average more than 153.79 miles. If H_0 is true (top panel), that happens 5% of the time: the false positive rate. Now suppose the new battery really averages 156 miles (bottom panel). The test misses it whenever the sample mean falls short of 153.79, which happens with probability $\beta = 0.0966$. The power, the chance of correctly detecting the better battery, is 0.9034.

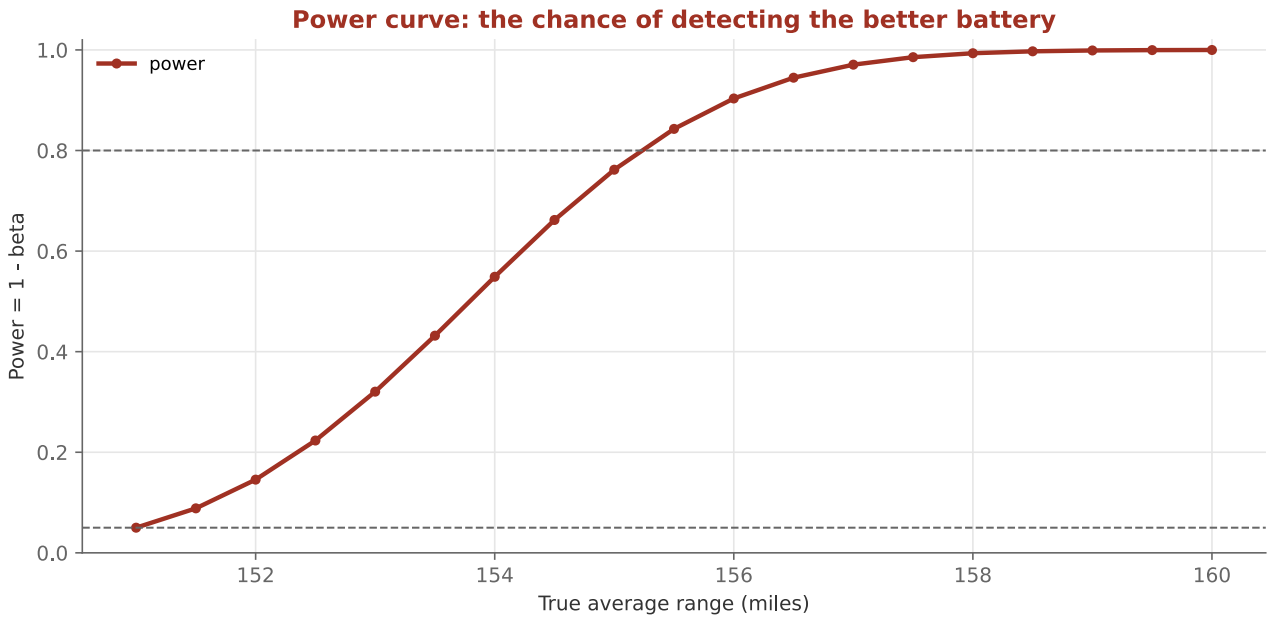
(The textbook gets 0.0968 and 0.9032 by rounding z to -1.30. Its version of the calculation is correct.)

We can check β the same way we checked the confidence level: test many fresh samples from a world where the mean really is 156, and count.

	value
samples	20000.000
rejected H0	18040.000
rejection rate	0.902
failed to reject (rate)	0.098

The simulated studies fail to reject 9.80% of the time, against the formula's 9.66%. The formula is not magic; it is a shortcut to what would happen if we ran the study over and over.

Power depends on how big the real effect is

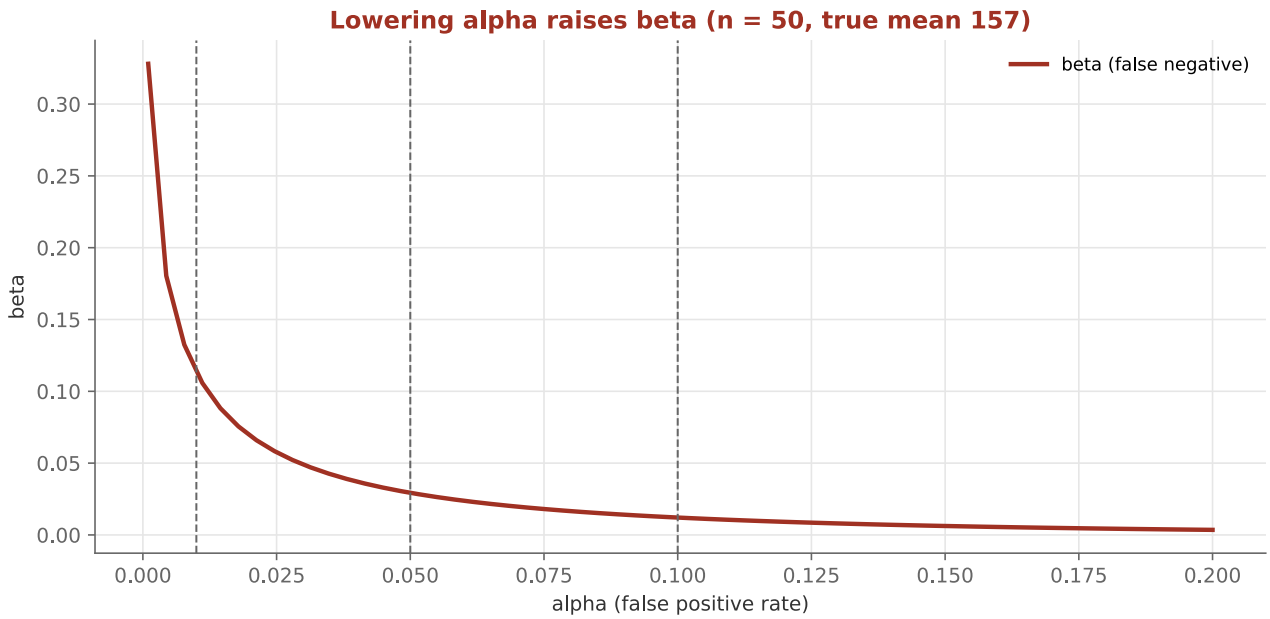


true mean	beta	power
151.0	0.9500	0.0500
152.0	0.8544	0.1456
153.0	0.6795	0.3205
154.0	0.4511	0.5489
155.0	0.2382	0.7618
156.0	0.0966	0.9034
157.0	0.0293	0.9707
158.0	0.0066	0.9934

If the true mean is barely above 151, the test almost never notices: power starts at $\alpha = 0.05$ (at exactly 151, rejecting *is* a Type I error) and is only 15% at 152 miles. Power passes 80% somewhere between 155 and 156. **A test can see large effects easily and small ones hardly at all.** Before trusting a "no significant difference," ask how large a difference the study could have detected.

(The textbook's Table 9.10 shows 0.0495 at 151, which should be exactly 0.05; it comes from rounding z_α to 1.65.)

Power depends on α



alpha	critical x-bar	beta	power
0.001	156.2443	0.3281	0.6719
0.005	155.3713	0.1686	0.8314
0.010	154.9479	0.1133	0.8867
0.025	154.3262	0.0576	0.9424
0.050	153.7914	0.0293	0.9707
0.100	153.1749	0.0121	0.9879
0.200	152.4283	0.0035	0.9965

At a true mean of 157 miles, cutting α from 0.05 to 0.01 raises β from 0.0293 to 0.1133. Fewer false positives, more false negatives: there is no free lunch.

The textbook reports $\beta = 0.1190$ for this case. It rounds $z_{0.01}$ to 2.33 and then rounds the critical mean to 155 miles before computing β; unrounded, the critical mean is 154.95 and $\beta = 0.1133$. The lesson is the same, but if you work in Excel you will match this note, not the book.

Power depends on the sample size, which you control

The textbook says the only way to reduce both errors at once is a larger sample. Here is how much larger. Suppose the improvement worth detecting is 3 miles (true mean 154):

n	beta	power
10	0.8035	0.1965
25	0.6535	0.3465

n	beta	power
50	0.4511	0.5489
75	0.3015	0.6985
100	0.1962	0.8038
150	0.0782	0.9218
200	0.0293	0.9707

With 50 cars, Nissan would detect a 3-mile improvement only 55% of the time, close to a coin flip. To reach the conventional 80% power it needs 99 cars, from

$$n = \left(\frac{(z_\alpha + z_\beta)\,\sigma}{\mu_1 - \mu_0} \right)^2 = \left(\frac{(1.645 + 0.842) \times 12}{3} \right)^2 .$$

This is the calculation to do **before** a study, alongside the margin-of-error calculation of Chapter 8. A study too small to detect the effect that matters is a study designed to say "no evidence," and it will usually oblige.

Explore this yourself in the interactive app below. Change the true mean, the hypothesized mean, σ, and n, and watch α, β, and the power move.

Launch the interactive simulation

Buick owners have averaged 59 years old. After an advertising campaign aimed at younger buyers, Buick will survey 41 new owners and test $H_0 : \mu \geq 59$ against $H_1 : \mu < 59$ at $\alpha = 0.05$, with $\sigma = 10$ years. **(a)** Find the critical sample mean. **(b)** If the campaign really lowered the average age to 58, what is the power? **(c)** Is this a well-designed study? How many owners would give 80% power against a true mean of 58?

i Solution

	value
mu0 (H0)	59
true mean	58
sigma	10
n	41
alpha	0.05
standard error	1.5617
critical x-bar	56.4312
beta = P(Type II error)	0.8424
power = 1 - beta	0.1576

(a) 56.43 years: Buick rejects H_0 if the sample averages less than that. **(b)** 0.1576. **(c)** No. The survey would detect a one-year drop only about 16% of the time. Reaching 80% power takes 619 owners. Either survey far more people or decide in advance that a one-year change is too small to matter.

Statistically significant is not the same as important

A result is **statistically significant** when $p \leq \alpha$: the data are unlikely under H_0. It is **practically significant** when the effect is large enough to matter for the decision. The first is a statistical question; the second is a business one.

	Target, all stores
x-bar (minutes)	4.95
n	20,000
p-value	0.0000003
95% interval: lower	4.930
95% interval: upper	4.970

Suppose Target rolls the kiosks out and logs 20,000 checkouts averaging 4.95 minutes. The p-value is about 0.0000003, overwhelming evidence that kiosks are faster than 5 minutes. By about **three seconds**. That is statistically significant and commercially irrelevant. The confidence interval shows it at once, which is another reason to report one.

The opposite also happens: a small study of an important effect, like the 41-owner Buick survey, can easily come back "not significant." **Large samples find trivial effects; small samples miss important ones.** Always ask *how big* as well as *whether*.

Excel at a Glance

step	Excel	note
z critical value, two-tailed	=NORM.S.INV(1 - alpha/2)	1.96 at alpha = 0.05
z critical value, one-tailed	=NORM.S.INV(1 - alpha)	1.645 at alpha = 0.05
t critical value, two-tailed	=T.INV.2T(alpha, df)	positive value
t critical value, one-tailed	=ABS(T.INV(alpha, df))	T.INV returns the LEFT-tail (negative) value
margin of error, sigma known	=CONFIDENCE.NORM(alpha, sigma, n)	z x sigma / SQRT(n)
margin of error, sigma unknown	=CONFIDENCE.T(alpha, s, n)	t x s / SQRT(n)
p-value, t, lower tail	=T.DIST(t, df, TRUE)	area to the left
p-value, t, upper tail	=T.DIST.RT(t, df)	area to the right
p-value, t, two-tailed	=T.DIST.2T(ABS(t), df)	needs a NON-NEGATIVE t
p-value, z, lower tail	=NORM.S.DIST(z, TRUE)	area to the left
p-value, z, upper tail	=1 - NORM.S.DIST(z, TRUE)	area to the right
p-value, z, two-tailed	=2*(1 - NORM.S.DIST(ABS(z), TRUE))	both tails
z-test from raw data	=Z.TEST(range, mu0, sigma)	UPPER-tail p-value only; lower tail is 1 - Z.TEST

Reading the Textbook: Corrections

Donnelly is a clear guide to the mechanics, and the reading for these chapters is assigned. A few statements in it will cost you points if you copy them:

1. **False positive / false negative (Section 9.5, Focus on Analytics).** The labels are reversed. A Type I error is a **false positive**; a Type II error is a **false negative**.
2. **"The p-value can also be described as the confidence you have that your null hypothesis is correct" (Section 9.2).** No. It is computed assuming H_0 is true and says nothing directly about the probability that H_0 is true.
3. **"The sample supports the original assumption."** After a failure to reject, the data are *consistent with* or *fail to contradict* H_0. They do not support it.
4. **"p-values can be computed directly only for z."** Only from the printed table. Excel's `T.DIST` family gives exact t p-values.
5. **Larger n helps "where the t-distribution can be approximated by the normal."** Larger n helps because the Central Limit Theorem makes $\bar{x}$ more normal. For skewed populations $n = 30$ may not be enough.
6. **Rounding.** `NORM.S.INV(0.95)` returns 1.645, not 1.654 as printed. The book's rounded z-values (2.33, 1.65) move some sample sizes by one and some β values in the second decimal place. Match Excel, and say so.

Which Procedure Is It?

For each, name the procedure (interval or test; mean with z or t, or proportion), write H_0 and H_1 where there is a test, and say whether it is one- or two-tailed.

(a) A casino host samples 45 high-limit players' average session length to estimate the mean for all such players.

(b) A hotel claims at least 90% of guests rate their stay 4 stars or better. A consumer group samples 300 reviews and wants evidence that the claim overstates satisfaction.

(c) A packaging line is supposed to fill 18-ounce boxes. Quality control samples 25 boxes an hour and wants to catch under-filling *or* over-filling.

(d) A marketing director wants to estimate the share of app users who would pay for a premium tier to within 3 points.

(e) A new call-routing system is adopted only if it cuts average hold time below the current 4.5 minutes. Twenty calls are sampled.

Solution

	Procedure	Hypotheses	Tails
(a)	t-interval for a mean	none	(two-sided interval)
(b)	z-test for a proportion, SE from $p_0 = 0.90$	$H_0 : p \geq 0.90$, $H_1 : p < 0.90$	one (lower)
(c)	t-test for a mean	$H_0 : \mu = 18$, $H_1 : \mu \neq 18$	two
(d)	sample size for a proportion, then a z-interval	none	(two-sided interval)
(e)	t-test for a mean (check skewness: $n = 20$ and hold times are skewed)	$H_0 : \mu \geq 4.5$, $H_1 : \mu < 4.5$	one (lower)

In (d), with no prior estimate use $p = 0.5$: 1,068 users at 95% confidence.

Laboratory 8–9

There are two ways to get these files to Claude. **Start with Path 1.** It works on every Claude account, including the free tier, and needs nothing turned on.

Path 1 — upload the files

Download the Laboratory 8–9 pack — one file. Unzip it and you will have a folder holding 9 files. Drag all 9 into a new Claude conversation, wait for each to show as attached, then paste this:

PROMPT

I have uploaded 9 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.

Then run `08_09_confidence_intervals_and_tests.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

If web access is working on your account, Claude can retrieve the files itself:

PROMPT

Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 8-9 and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.

Then run `08_09_confidence_intervals_and_tests.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach the page, **nothing is wrong with your work or with the repository.** Claude's web access is a setting, and even when it is on, Claude cannot always retrieve a file by direct link. Use Path 1 instead — it is the ordinary route, not a consolation prize.

Your own junior analyst

The lab script answers the questions in this note. The skill to practice is getting a *new* question answered and checking the answer:

PROMPT

A Las Vegas restaurant group says its average ticket is $62. I sampled 40 tickets last weekend: the mean was $67.40 and the standard deviation was $21.30. By running code, (1) give me a 95% confidence interval for the average ticket, (2) test whether the average differs from $62 at alpha = 0.05 and give the exact p-value, and (3) tell me whether the two answers agree and why they must. Then state, in one sentence each, what a Type I error and a Type II error would mean here, and label each as a false positive or a false negative. Finally, tell me whether n = 40 is large enough to trust a t-procedure for restaurant tickets, and how you would check.

Check four things in the reply: the t critical value has 39 degrees of freedom; the p-value is two-tailed; the Type I error is labelled a **false positive** (concluding the average differs when it does not); and the answer to the last question is not simply "yes, because 40 is more than 30."

Working in Excel

PROMPT

Build me an Excel workbook that shows what "95% confidence" means. Sheet 1 has input cells for a true mean (100), a true standard deviation (16), a sample size n (30), and a confidence level (0.95). Sheet 2 draws 100 samples of size n with =NORM.INV(RAND(), mean, sd), and for each computes x-bar, s, the margin of error with CONFIDENCE.T, the lower and upper limits, and a column that says TRUE if the interval contains the true mean. Report the share of TRUE at the top, and chart the 100 intervals as horizontal error bars against a vertical line at the true mean. Give me the file to download.

Press F9 a dozen times and note the share each time. Then change the confidence level to 0.80 and repeat. In one paragraph: what changed, what did not, and what does that tell you about the meaning of the word "confidence"?